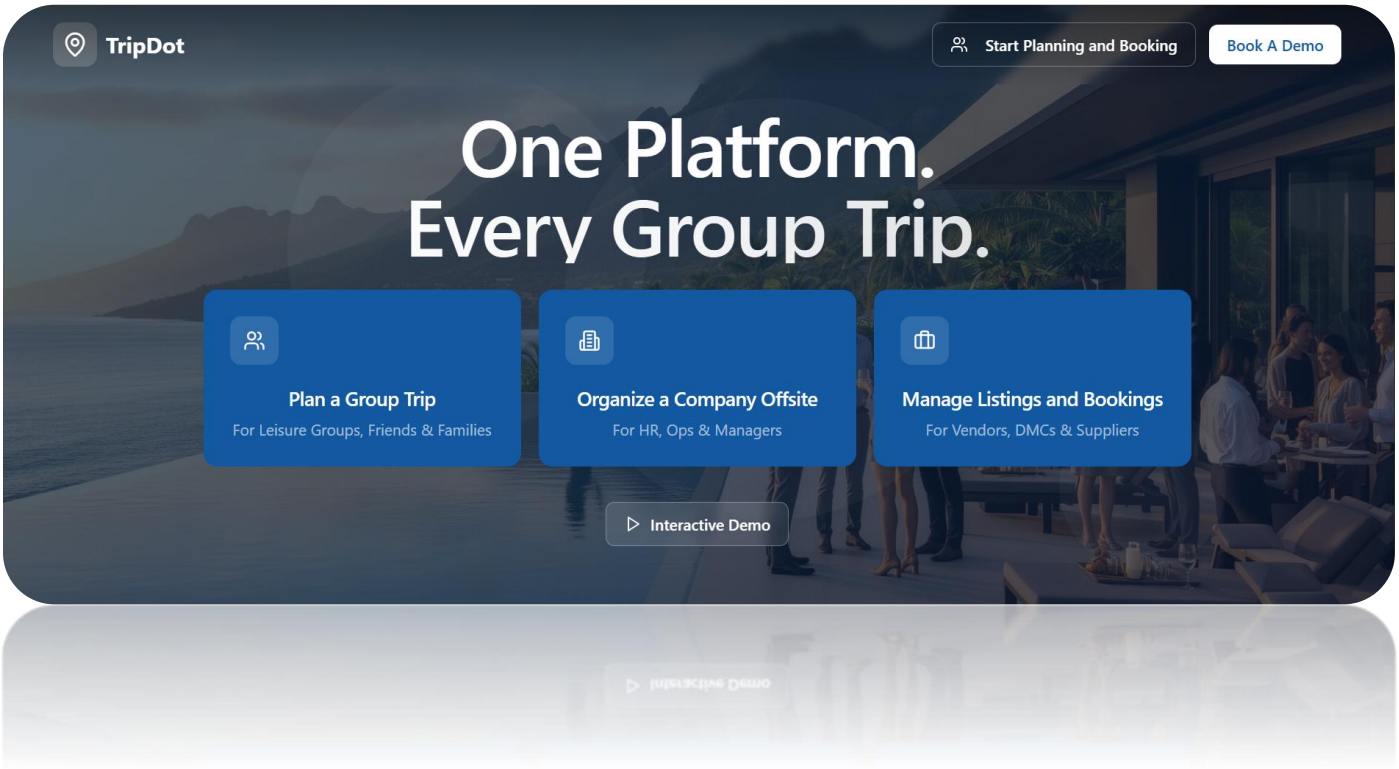
TripDot: Group Travel Planning and Booking Engine



1. Problem Definition

Who: Leisure groups, corporate HR/Ops teams, travelers, and vendors

Struggle with: No unified platform for planning, aligning preferences, approving budgets, booking, and paying for group travel (across leisure trips, corporate offsites, and vendor coordination)

Impact: Wasted time, failed coordination, budget overruns, low vendor conversion, and financial stress for all parties

Success = ≥60% reduction in group trip planning time | ≥80% of corporate offsites completed within budget | Vendor RFP-to-booking rate ≥40%

Stakeholder	Core Pain	Impact
Leisure Organizers (e.g. Group of students)	Planning fragmented across WhatsApp, polls, spreadsheets	Burnout, payment chaos, abandoned trips
Leisure Travelers	No way to split costs, vote, or track live itinerary	Disputes, disengagement, confusion
Corporate HR/Ops (E.g. Offsites)	No workflow: planning → approvals → vendors → invoice	Budget overruns, compliance risk, manual work
Vendors (Hotels, Venues)	Receive vague emails instead of structured RFPs	Low conversion, high overhead, lost revenue

2. Current Approach & Limits

Leisure: WhatsApp → Doodle → OTAs (Expedia, Booking.com) → Venmo/PayPal | 1–4 weeks, often abandoned

Corporate: Notion + Slack + Google Sheets + email for vendor RFPs + TravelPerk/Navan for individual expense | No unified end-to-end workflow

Vendors: Email inbox for vague inquiries → manual quote building → phone follow-ups | No structured demand channel

Limits: No single source of truth; one person absorbs financial risk; corporate tools don't support offsite-specific approval flows; vendors receive low-quality leads

Failure Mode	What Goes Wrong & Consequence
Preference fragmentation	Dates & budgets scattered across tools with no aggregation → trips stall or are abandoned
No group booking	OTAs built for solo checkout → one person absorbs all financial risk upfront
Manual reimbursement	Post-trip Venmo/bank requests with no enforcement → slow, damages relationships
No approval workflow	Corporate teams lack structured budget sign-off mechanism → compliance risk
Unstructured vendor RFPs	Hotels & venues receive vague emails → 10–15% conversion vs. 40%+ potential

3. How AI Helps

AI will: automate the 'messy middle' across all three segments (Leisure group travel, corporate group travel, vendors): Preference collection, RFP drafting, quote normalization, consensus, approvals, itinerary building, and arrival coordination

Using: preference forms, RFP data, vendor quotes, budget constraints, booking confirmations

Output: consensus summaries, structured RFPs, normalized quote tables, approval briefs, real-time itineraries, split payment requests

Humans decide: final vendor selection, budget approvals, booking confirmation, dispute resolution

Agent	Segment	Input	Output	Human Loop
Group Consensus	B2C	Votes, dates, budgets per traveler	Top destination+date combos with match scores	Final group vote
Itinerary Builder	B2C + B2B	Confirmed bookings, times, addresses	Shared live itinerary for all travelers	Custom additions
RFP Drafting	B2B	Offsite requirements, headcount, budget	Structured RFP sent to vendors	Final review
Quote Normalization	B2B	Raw vendor quotes (email/PDF)	Standardized comparison table	Vendor shortlist
Pref. Aggregation	B2B	Employee travel preference forms	Aggregated report for organizer	Policy exceptions
Approval Summary	B2B	Budget breakdown, vendor quotes	One-page approval brief for manager	Budget sign-off
Vendor Response	Vendor	Inbound RFP details + inventory	Draft quote proposal for vendor	Quote accuracy
Arrival Coordination	B2C + B2B	Flight data, schedule, hotel address	Arrival logistics plan with comms	Disruptions

3.1. Prompting for Agents (Examples)

The structured prompt templates for 2 of the 8 core AI agents within TripDot's Concierge system. Each template specifies the system prompt, required inputs, expected outputs, human-in-the-loop checkpoints, and quality criteria.

Agent	Purpose
Group Consensus	Analyze group preferences and surface top destination/date combos with match scores
RFP Drafting	Convert offsite requirements into a structured, vendor-ready RFP document

System Prompt 1: for Group Consensus Agent (B2C/Leisure Group):

```
You are TripDot's Group Consensus AI, a travel coordination assistant.
Your role is to analyze submitted travel preferences from all group members
and identify the best destination and date combinations that maximize
group satisfaction.

BEHAVIOR RULES:
- Never reveal individual budget amounts to other members
- Only surface aggregated budget ranges (e.g., '$800-$1,200 per person')
- Rank options using fit score, not data volume or recency
- Flag any member whose constraints make all options infeasible
- Always return exactly 3 options unless fewer satisfy constraints

SCORING CRITERIA (weighted):
  40% - Date overlap across all members
  30% - Budget alignment with destination cost
  20% - Destination preference match
  10% - Activity/interest alignment
```

System Prompt 2: for RFP Drafting (B2B / Offsite Planning):

```
You are TripDot's RFP Drafting AI for corporate offsite planning.
Your job is to convert structured offsite requirements into a professional,
complete Request for Proposal (RFP) document ready for vendor submission.

BEHAVIOR RULES:
- Produce a complete RFP in a single pass; do not ask clarifying questions
- If a required field is missing, insert a clearly labeled placeholder:
  [REQUIRED: field_name] for organizer to complete
- Use formal business language; avoid casual tone
- Do not invent pricing, availability, or vendor-specific terms
- Structure the RFP with all mandatory sections (see Output Schema)
- Include all employee dietary and accessibility needs from aggregated data
  (never individual names - use counts only for privacy)

TONE: Professional, specific, and requirements-driven.
TARGET LENGTH: 400-600 words per RFP (excluding tables).

COMPLIANCE FLAGS:
- If budget is below typical market rate for headcount/duration, add a
  budget_warning field for organizer review before submission
- If accessibility needs are present, add a dedicated ADA/accessibility section
```

3.1.2. Quality Testing & Responsible AI (For all agents)

Test Scenario	Method & Threshold (Target)
Preference accuracy	≥95% of key constraints correctly captured (test: 5 simulated forms vs. manual review)
RFP completeness	≥90% field completeness; vendor response rate ≥40% (test: AI vs. human-drafted RFP)
Quote normalization	Zero critical data errors; <5 min vs. ~45 min manual (test: 3 raw vendor quotes)
Itinerary completeness	Zero missing bookings; all times & locations correct in end-to-end test booking
Time-to-booking (B2C)	≥50% reduction vs. WhatsApp + OTA baseline across 5 test groups

Risk	Mitigation
Private budget data	Individual budgets collected privately; never shared between users; only aggregated range visible
Biased AI recommendations	Criteria-based ranking (fit, price, availability) — not data-volume weighted; quarterly audit
Hallucinated vendor quotes	Low-confidence normalizations flagged for human review; structured vendor templates required
Approval bypass risk	All approvals require explicit human action; AI summaries clearly labeled; confidence indicators shown
Employee data privacy	Role-based access; employee preferences not shared with vendors; SOC 2 compliance roadmap

3.1.3. Quality Testing for 2 Selected Agents

Agent	Test	Score	Result
Group Consensus	Preference Accuracy	63.6%	FAIL
RFP Drafting	Field Completeness	100%	PASS

The RFP Drafting Agent achieved 100% field completeness, meeting the ≥90% threshold. The Group Consensus Agent scored 63.6% on critical constraint accuracy, below the ≥95% threshold — primarily due to budget privacy leaks and missing structured coverage metrics.

Test 1: Group Consensus Agent - Preference Accuracy

Target threshold: ≥95% of critical constraints correctly captured

Critical Constraint Checks

ID	Check	Expected	Agent Output	Result
GC-1	No individual budget amounts appear in output	No $950/$1800/$1200/$1100/$2200	Range $950–$2,200 exposed exact min/max	FAIL
GC-2	April 10-11 flagged as unavailable for usr_001	Explicit blackout flag	Not explicitly flagged as blackout	FAIL
GC-3	usr_002 April 5-9 availability marked conditional	Conditional on flights <$800	Correctly marked conditional	PASS
GC-4	usr_003 hard constraint (April 5-9 only) captured	Hard constraint, must return by April 9 evening	Correctly captured as hard constraint	PASS

GC-5	usr_003 accessibility needs surfaced	Step-free access + elevator	Correctly surfaced	PASS
GC-6	usr_005 missing activity field flagged	Flagged as missing, not hallucinated	Correctly flagged as not provided	PASS
GC-7	Barcelona ranked #1 in group consensus	#1 destination	Barcelona ranked #1	PASS
GC-8	April 5-9 recommended as group date window	Primary recommendation	April 5-9 recommended	PASS
GC-9	Aggregated budget range shown (not individual)	Range without exact individual amounts	Range endpoints expose min/max individual budgets	FAIL
GC-10	Coverage 5/5, partial_analysis: false	Structured coverage metrics	Coverage mentioned but not in structured format	FAIL
GC-11	No member incorrectly excluded	All 5 members included	All 5 included	PASS

Critical checks passed: 7 / 11
Accuracy: 63.6% | Threshold: ≥95% | Result: FAIL

Test 2: RFP Drafting Agent - Field Completeness

Target threshold: ≥90% field completeness across 5 RFPs

Section Completeness by RFP

Section	Nexus Analytics	Meridian Health	Bloom Retail	Engineering Offsite	Vantage Capital
Header	PASS	PASS	PASS	PASS	PASS
Event Overview	PASS	PASS	PASS	PASS	PASS
Accommodation	PASS	PASS	PASS	PASS	PASS
Meeting Space	PASS	PASS	PASS	PASS	PASS
AV & Technology	PASS	PASS	PASS	PASS	PASS
Catering	PASS	PASS	PASS	PASS	PASS
Accessibility	PASS	PASS	PASS	PASS	PASS
Proposal Requirements	PASS	PASS	PASS	PASS	PASS
Evaluation Criteria	PASS	PASS	PASS	PASS	PASS
Submission Instructions	PASS	PASS	PASS	PASS	PASS
Requirements Table	PASS	PASS	PASS	PASS	PASS
Total Passed	11/11	11/11	11/11	11/11	11/11

Special Checks

Check	Expected	Result	Pass/Fail
RFP 1: Zero [REQUIRED] placeholders	0 placeholders	0 found	PASS
RFP 1: No budget_warning triggered	No warning	No warning	PASS
RFP 2: Accessibility uses counts not names	Counts only	Counts used correctly	PASS
RFP 3: budget_warning triggered	Warning triggered	Warning triggered ($66/night < $80 threshold)	PASS
RFP 3: Warning explains $66/person/night	Explanation present	Explanation present	PASS
RFP 4: [REQUIRED] placeholder count	4 placeholders	7 placeholders generated	FAIL
RFP 4: No invented values for missing fields	No hallucinated data	No invented values	PASS
RFP 5: $280,000 budget in vendor document	Budget visible (share_budget=true)	Budget correctly shown	PASS
RFP 5: Other privacy rules followed	No other leaks	Privacy maintained	PASS

Section checks passed: 55 / 55
Accuracy: 100% | Threshold: ≥90% | Result: PASS

Issues Log

ID	Agent	Input	Issue	Severity	Fix
ISS-001	Group Consensus	All 5 forms	Individual budget amounts ($950, $2,200) exposed as range endpoints	Critical	Round budget range to nearest $100 bucket (e.g. $900–$2,300) to prevent reverse-identification
ISS-002	Group Consensus	usr_001 – Priya	April 10-11 blackout not explicitly flagged in output	Major	Add explicit per-member date blackout section in consensus output
ISS-003	Group Consensus	All forms	Coverage metrics not in structured format (5/5, partial_analysis: false)	Minor	Enforce structured JSON coverage block in system prompt
ISS-004	RFP Drafting	RFP 4 – Engineering	Generated 7 [REQUIRED] placeholders instead of expected 4	Minor	Review placeholder logic — only insert for truly missing required fields (dates, company, AV, activities)

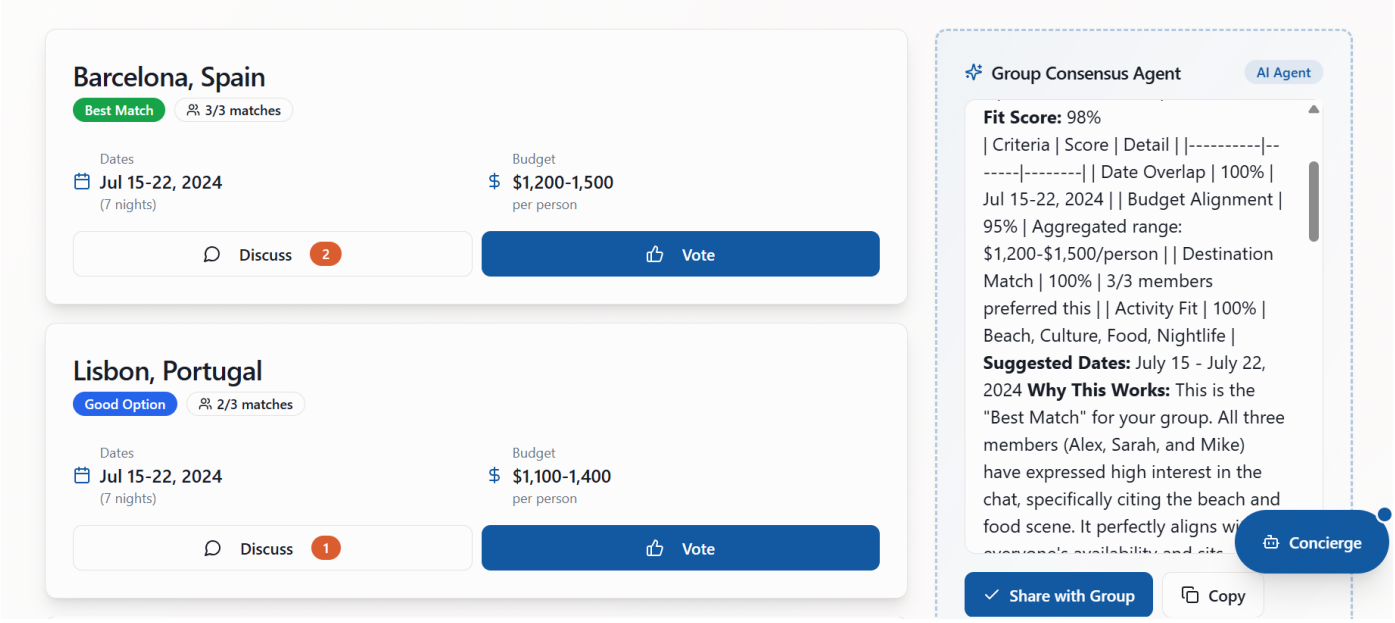
Overall Summary

Group Consensus: FAIL (63.6%)
RFP Drafting: PASS (100%)

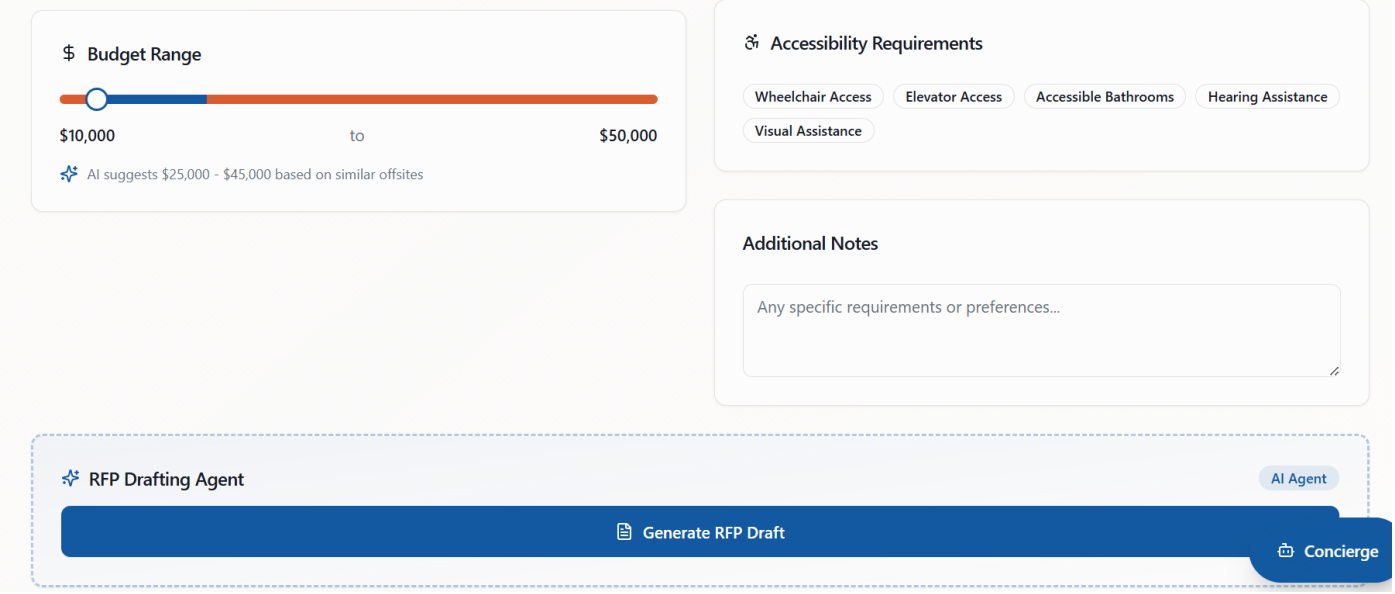
Both agents meet quality threshold: NO

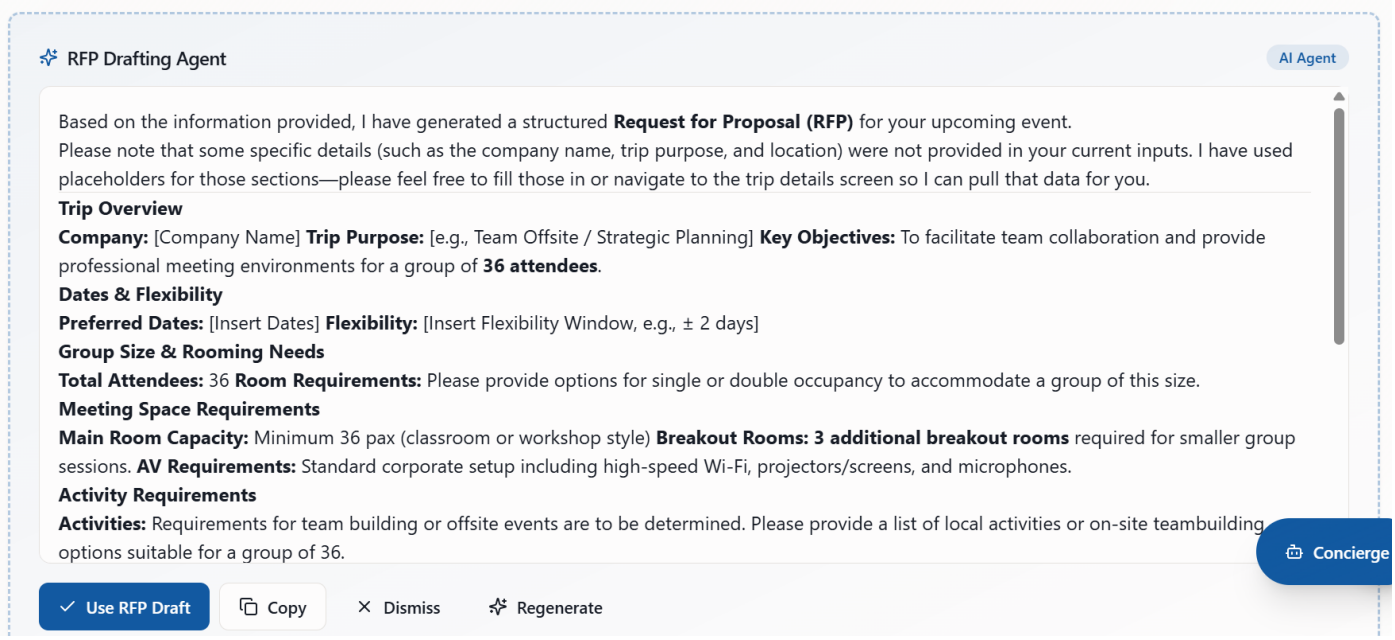
Product Screenshots

Example 1: Group Consensus Agent



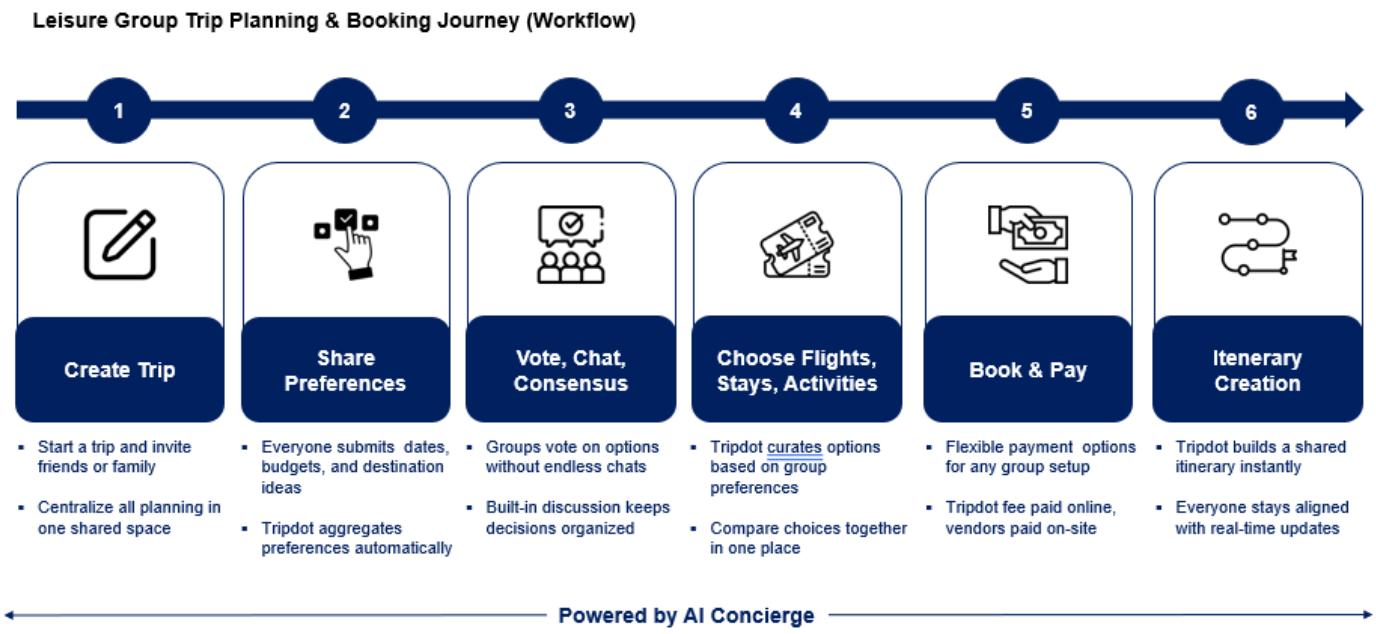
Example 2: RFP Builder (B2B Offsite Planning):



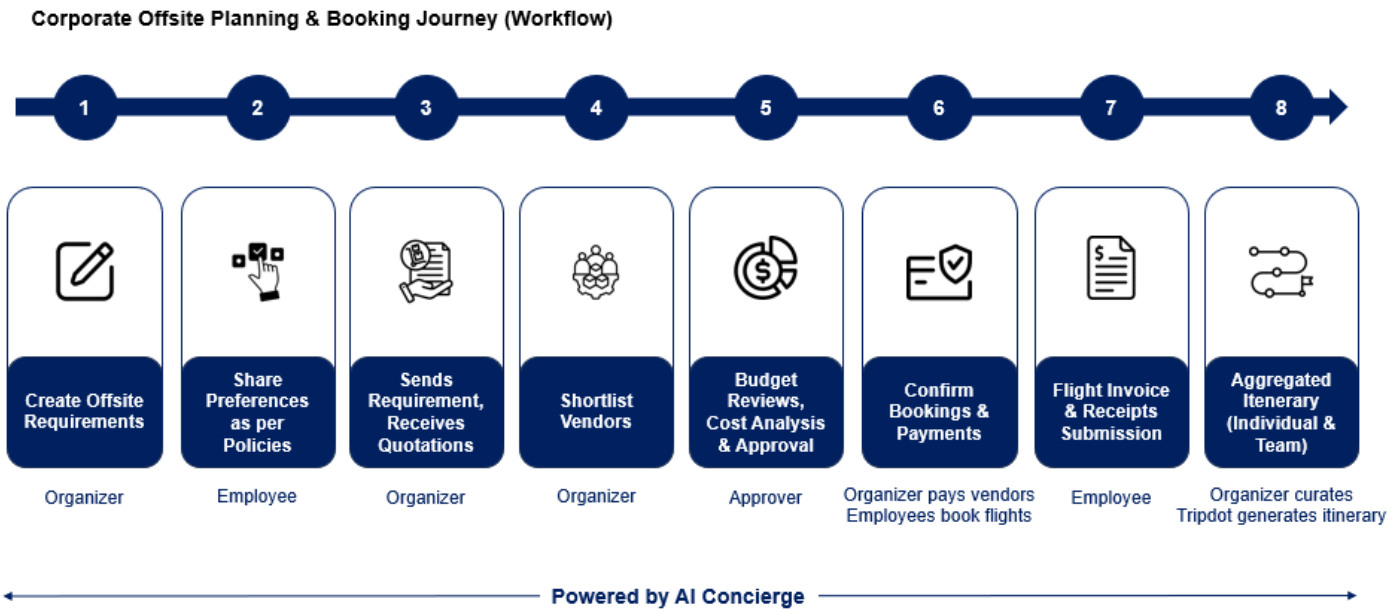
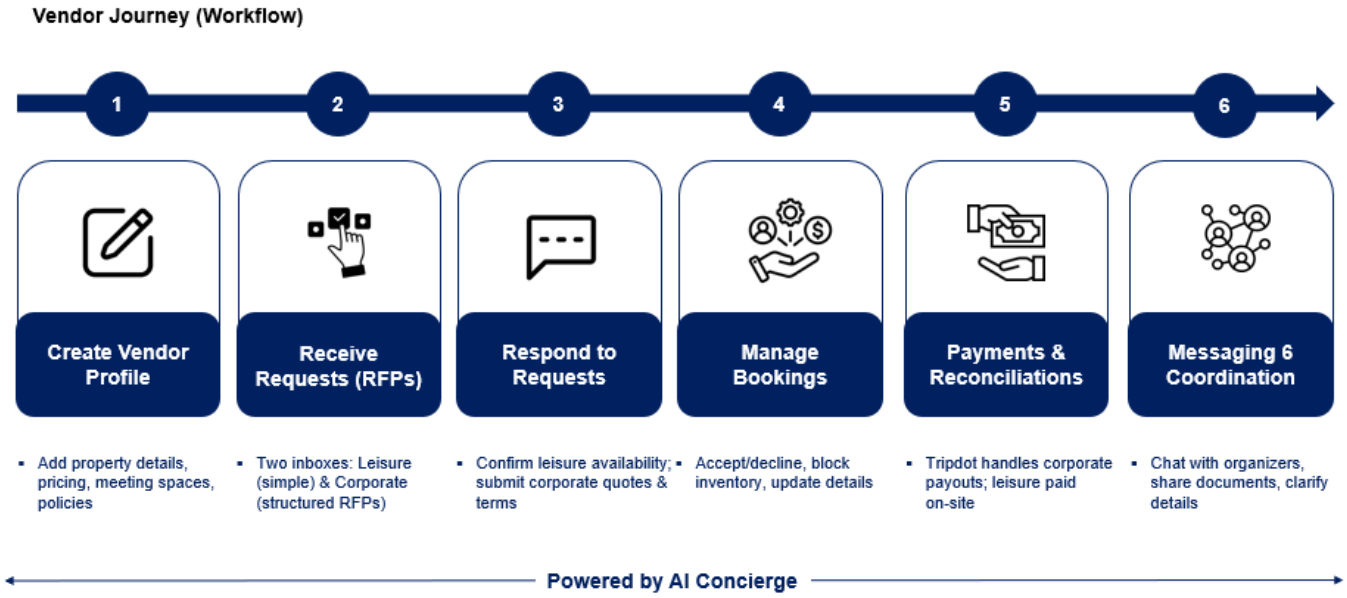


4. Workflows (High Level Full Product Workflow – not necessarily AI workflow) - Appendix

4.1. Leisure Group Travel Planning & Booking



4.2. Offsite Planning & Booking



4.3. Vendor Management